

EXPERIMENT- 02

Python Code:

```
02.py > ...
1  alphabet = "abcdefghijklmnopqrstuvwxyz"
2
3  plaintext = input("Enter the text: ")
4  key = input("Enter the 26-letter key: ").lower()
5
6  if len(set(key)) == 26 and len(key) == 26 and key.isalpha():
7      encrypted_text = ""
8      for char in plaintext:
9          if char.isalpha():
10             is_upper = char.isupper()
11             index = alphabet.index(char.lower())
12             cipher_char = key[index]
13             encrypted_text += cipher_char.upper() if is_upper else cipher_char
14          else:
15             encrypted_text += char
16
17     print("Encrypted Text:", encrypted_text)
18
19     decrypted_text = ""
20     for char in encrypted_text:
21         if char.isalpha():
22             is_upper = char.isupper()
23             index = key.index(char.lower())
24             plain_char = alphabet[index]
25             decrypted_text += plain_char.upper() if is_upper else plain_char
26         else:
27             decrypted_text += char
28
29     print("Decrypted Text:", decrypted_text)
30
31 else:
32     print("Invalid key! Please provide exactly 26 unique alphabetic characters.")
```

Output:

```
PS D:\College\Crypto\Experiments> python -u "d:\College\Crypto\Experiments\02.py"
Enter the text: Hello World!
Enter the 26-letter key: QWERTYUIOPASDFGHJKLZXCVBNM
Encrypted Text: Itssg Vgksr!
Decrypted Text: Hello World!
PS D:\College\Crypto\Experiments> |
```